Generation of stable cell lines using DNA transposases

Risk assessment

– Work with human-derived material or transgenic cell lines (BSL-2)

▷ Wear gloves, safety glasses, lab coat

□ Dispose waste after inactivation as REGULATED MEDICAL WASTE

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This is a bench card. Full protocol available online.

Transfection of host cells

(1.) Seed cells to reach 70% confluence (50 000 cells per well in a 6-well plate) at the time of transfection.

(2.) Prepare a mix of the transposase and transfer plasmids at a sub-stoichiometric ratio.

Plasmid		Per well	
Transposase plasmid	Super PiggyBac transposase (SBI)	0.05 pmol	0.32 µg
Transfer plasmid	Custom	0.20 pmol	1.00 µg

(3.) Transfect the cells using a non-viral method such as Lipofectamine® SOP 0012.

(4.) After 24 h, replace the medium.

(5.) Begin selection or screening for stably transfected cells according to SOP 0019.

Critical: The Limited Use Research License (LURL) of the piggyBac transposase (System Biosciences/Demeetra AgBio) applies to all derived cell lines, persists until those lines are destroyed, and excludes any transfer to, use by, or contracted work for industry or commercial third parties.

Notes (available online)

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